Personalized Antidepressant Treatment Recommendation Using Reinforcement Learning and Predictive Modelling on Synthetic Patient Data

R. de Filippis, Abdullah Al Foysal

Journal:	Publication Year:	Citations:	DOI:
OALib	2025.0	0	10.4236/oalib.1113957

CLINICAL & AGENTIC AI TAKEAWAY

This study developed a clinical decision support system that personalizes antidepressant treatment selection using synthetic patient data, predictive modeling, and reinforcement learning, aiming to reduce patient suffering and improve treatment success.

ABSTRACT

This study presents a comprehensive clinical decision support system aimed at personalizing antidepressant treatment selection using synthetic patient data, predictive modelling, and reinforcement learning.Traditional antidepressant prescribing often relies on trial-and-error methods, which can result in prolonged patient suffering and delayed treatment success, particularly in treatment-resistant depression.To address the limitations posed by data scarcity and privacy constraints in psychiatric research, we developed a robust synthetic patient data generator that simulates realistic clinical, genetic, and treatment response profiles.These synthetic datasets capture essential features such as age, gender, baseline depression severity, genetic markers (5HTTLPR, BDNF, COMT, FKBP5), treatment history, and comorbidities including anxiety and chronic pain.We employed Gradient Boosting and Neural Network classifiers to predict treatment response probabilities based on individual patient profiles.These models achieved moderate classification accuracy, with a tendency to predict responders more reliably than non-responders.To further optimize treatment strategies, we integrated a Dueling DQN Reinforcement Learning agent trained within a custom-designed environment that simulates multi-step treatment processes, side effect profiles, and severity progression.While the reinforcement learning agent successfully optimized sequential treatment selections and reduced symptom severity, it did not achieve remission under the current reward configuration, suggesting the need for further reward function tuning.An enhanced clinical decision support system was developed to generate top treatment recommendations with natural language explanations, facilitating transparent and interpretable clinical guidance.This research demonstrates the po-How to cite this paper: de

Study Overview & Background

This study presents a comprehensive clinical decision support system aimed at personalizing antidepressant treatment selection using synthetic patient data, predictive modelling, and reinforcement learning.Traditional antidepressant prescribing often relies on trial-and-error methods, which can result in prolonged patient suffering and delayed treatment success, particularly in treatment-resistant depression.To address the limitations posed by data scarcity and privacy constraints in psychiatric research, we developed a robust synthetic patient data generator that simulates realistic clinical, genetic, and treatment response profiles.These synthetic datasets capture essential features such as age, gender, baseline

depression severity, genetic markers (5HTTLPR, BDNF, COMT, FKBP5), treatment history, and comorbidities including anxiety and chronic pain.We employed Gradient Boosting and Neural Network classifiers to predict treatment response probabilities based on individual patient profiles.These models achieved moderate classification accuracy, with a tendency to predict responders more reliably than non-responders.To further optimize treatment strategies, we integrated a Dueling DQN Reinforcement Learning agent trained within a custom-designed environment that simulates multi-step treatment processes, side effect profiles, and severity progression.While the reinforcement learning agent successfully optimized sequential treatment selections and reduced symptom severity, it did not achieve remission under the current reward configuration, suggesting the need for further reward function tuning.An enhanced clinical decision support system was developed to generate top treatment recommendations with natural language explanations, facilitating transparent and interpretable clinical guidance.This research demonstrates the po-How to cite this paper: de

Methodology & Clinical Focus

Investigation into artificial intelligence models, machine learning algorithms, and clinical predictive tools applied to orthopedic surgery and musculoskeletal medicine.

Key Findings & Analysis

This study developed a clinical decision support system that personalizes antidepressant treatment selection using synthetic patient data, predictive modeling, and reinforcement learning, aiming to reduce patient suffering and improve treatment success.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** R. de Filippis, Abdullah Al Foysal
- **Journal / Venue:** OALib (2025.0)
- **SJR Metric:** Quartile N/A
- **Citation Count:** 0 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.4236/oalib.1113957
- **Consensus Evidence Link:** https://consensus.app/papers/personalized-antidepressant-treatment-recommendation-filippis-foysal/300781caf22a571891d5feb3f8306cd5/

Source & Provenance: Curated for Agentic Orthopedics Research Collection • DOI: https://doi.org/10.4236/oalib.1113957 • Consensus Page: https://consensus.app/papers/personalized-antidepressant-treatment-recommendation-filippis-foysal/300781caf22a571891d5feb3f8306cd5/ • Verified and rendered via Deep Search & Academic Repositories.